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Semiconductor package

Abstract

A semiconductor package includes a first semiconductor chip on a base chip, a second semiconductor chip on the first semiconductor chip in a first direction, each of the first and second semiconductor chips including a TSV and being electrically connected to each other via the TSV, dam structures on the base chip and surrounding a periphery of the first semiconductor chip, a first adhesive film between the base chip and the first semiconductor chip, a portion of the first adhesive film filling a space between the first semiconductor chip and the dam structures, a second adhesive film between the first semiconductor chip and the second semiconductor chip, a portion of the second adhesive film overlapping the dam structures in the first direction, and an encapsulant encapsulating a portion of each of the dam structures, the first semiconductor chip, and the second semiconductor chip.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0108325, filed on Aug. 17, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.
BACKGROUND
1. Field
(2) The present disclosure relates to a semiconductor package.
2. Description of the Related Art
(3) With weight reductions and performance improvements in electronic devices, miniaturization and high performance are also required in a field of semiconductor packages. To implement miniaturization, weight reduction, high performance, high capacity, and high reliability in a semiconductor package, research and development (R&D) of a semiconductor package having a structure, in which semiconductor chips are stacked in multiple stages, is continuously being conducted.
SUMMARY
(4) According to an example embodiment, a semiconductor package includes a base chip; a first semiconductor chip and second semiconductor chips sequentially stacked on the base chip in a first direction, including a through-silicon via (TSV), and electrically connected to each other by the through-silicon via; dam structures disposed on the base chip and surrounding a periphery of the first semiconductor chip; a first adhesive film disposed between the base chip and the first semiconductor chip; second adhesive films disposed between the first semiconductor chip and the second semiconductor chips; and an encapsulant encapsulating at least a portion of each of the dam structures, the first semiconductor chip, and the second semiconductor chips. At least a portion of the first adhesive film fills spaces between the first semiconductor chip and the dam structures, and at least portions of the second adhesive films overlap the dam structures in the first direction.
(5) According to an example embodiment, a semiconductor package includes a base chip; a semiconductor chip disposed on the base chip; dam structures disposed on the base chip and surrounding a side surface of the semiconductor chip; and an adhesive film disposed between the base chip and the semiconductor chip and protruding further the side surface of the semiconductor chip to be in contact with an inner surface of the dam structures. An upper surface of the dam structures is disposed on a level lower than a level of an upper surface of the semiconductor chip.
(6) According to an example embodiment, a semiconductor package includes a base chip; a first semiconductor chip and second semiconductor chips sequentially stacked on the base chip; dam structures disposed on the base chip to respectively correspond to side surfaces of the first semiconductor chip; a first adhesive film disposed between the base chip and the first semiconductor chip; and second adhesive films disposed between the first semiconductor chip and a lowermost second semiconductor chip among the second semiconductor chips, and between the

second semiconductor chips. The first adhesive film or a lowermost second adhesive film among the second adhesive films is in contact with an upper surface of the dam structures.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Features will become apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings, in which:

(2) FIG. **1**A is a cross-sectional view of a semiconductor package according to an example embodiment.

(3) FIG. **1**B is a partially enlarged view of region "A" of FIG. **1**A.

(4) FIG. **1**C is a plan view of the semiconductor package of FIG. **1**A.

(5) FIG. **2** is a partially enlarged view illustrating a modified example of the semiconductor package of FIG. **1**A.

(6) FIG. **3** is a plan view illustrating a modified example of the semiconductor package of FIG. **1**A.

(7) FIG. **4**A is a cross-sectional view of a semiconductor package according to an example embodiment.

(8) FIG. **4**B is a partially enlarged view of region "B" of FIG. **4**A.

(9) FIG. **5**A is a plan view of a semiconductor package according to an example embodiment.

(10) FIG. **5**B is a cross-sectional view taken along line I-I′ of FIG. **5**A.

(11) FIG. **6** is a cross-sectional view of a semiconductor package according to an example embodiment.

(12) FIGS. **7**A to **7**I are cross-sectional views of stages in a method of manufacturing a semiconductor package according to an example embodiment.

DETAILED DESCRIPTION

(13) FIG. **1**A is a cross-sectional view of a semiconductor package **1000**A according to an example embodiment, FIG. **1**B is a partially enlarged view of region "A" of FIG. **1**A, and FIG. **1**C is a plan view of the semiconductor package **1000**A of FIG. **1**A.

(14) Referring to FIGS. **1**A to **1**C, a semiconductor package **1000**A according to an example embodiment may include a base chip **100**, a semiconductor chip **200**, an adhesive film **300**, and a dam structure **500**. The semiconductor package **1000**A may further include an encapsulant **400** encapsulating the semiconductor chip **200**.

(15) In a thermocompression bonding process of fixing the semiconductor chip **200** to the base chip **100**, an extending region of an adhesive film formed to protrude farther than a side surface of the semiconductor chip **200** may cause a poor exterior and a decrease in reliability of a semiconductor package. In contrast, in the present disclosure, an extending region of the adhesive film **300** between the base chip **100** and the semiconductor chip **200** may be controlled using the dam structure **500** to be prevented from protruding outwardly of the base chip **100**, and thus, reliability of the semiconductor package **1000**A may be improved.

(16) For example, the base chip **100** may include a semiconductor material, e.g., a silicon (Si) wafer. In another example, the base chip **100** may be a printed circuit board (PCB) or a glass substrate which does not include a semiconductor material. In an example embodiment, the base chip **100** may include a substrate **101**, an upper passivation layer **103**, an upper pad **105** and a lower pad **104**, a device layer **110**, an external connection terminal **120**, and a through-silicon via (TSV) **130**. However, when the base chip **100** is a PCB or a glass substrate which does not include a semiconductor material, the base chip **100** may not include a device layer and a TSV.

(17) For example, the base chip **100** may be a buffer chip including a plurality of logic devices and/or memory devices in the device layer **110**. Accordingly, the base chip **100** may transmit a signal from the semiconductor chip **200** stacked thereon to an external entity, and may also transmit

a signal and power from an external entity to the semiconductor chip **200**. For example, the base chip **100** may perform both logic functions and memory functions through logic devices and memory devices. In another example, the base chip **100** may include only logic devices to perform only logic functions.

(18) For example, the substrate **101** may include a semiconductor element, e.g., silicon or germanium (Ge), or a compound semiconductor, e.g., silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), or indium phosphide (InP). In another example, the substrate **101** may have a silicon-on-insulator (SOI) structure. The substrate **101** may include a conductive region, e.g., a well doped with impurities or a structure doped with impurities. The substrate **101** may include various device isolation structures, e.g., a shallow trench isolation (STI) structure.

(19) The upper passivation layer **103** may be formed on an upper surface of the substrate **101** and may protect the substrate **101**. For example, the upper passivation layer **103** may be formed of silicon, silicon nitride, or silicon oxynitride film. In another example, the upper passivation layer **103** may be formed of a polymer, e.g., polyimide (PI). A lower passivation layer may be further formed on a lower surface of the device layer **110**.

(20) The upper pad **105** may be disposed on the upper passivation layer **103**. The upper pad **105** may include at least one of, e.g., aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au). The lower pad **104** may be disposed below the device layer **110**, and may include a material similar to that of the upper pad **105**.

(21) The device layer **110** may be disposed on a lower surface of the substrate **101** and may include various types of devices. For example, the device layer **110** may include a field-effect transistor (FET), e.g., a planar FET or a FinFET, a memory device, e.g., a flash memory, a dynamic random access memory (DRAM), a static random access memory (SRAM), an electrically erasable programmable read-out (EEPROM), a phase-change random access memory (PRAM), a magnetoresistive random access memory (MRAM), a ferroelectric random access memory (FeRAM), a resistive random access memory (RRAM), or the like, a logic device, e.g., an AND, OR, NOT, and the like, and various active and/or passive devices, e.g., a system large scale integration (LSI), a CMOS Imaging Sensor (CIS), and a micro-electro-mechanical system (MEMS).

(22) The device layer **110** may include an interlayer insulating layer **111** and a multilayer interconnection layer **112** on the above-described devices. The interlayer insulating layer **111** may include, e.g., silicon oxide or silicon nitride. The multilayer interconnection layer **112** may include a multilayer interconnection and/or a vertical contact. The multilayer interconnection layer **112** may connect devices of the device layer **110** to each other, devices to a conductive region of the substrate **101**, or devices to an external connection terminal **120**.

(23) The external connection terminal **120** may be disposed on the lower pad **104**, and may be connected to the interconnection layer **112** or the TSV **130** inside the device layer **110**. For example, the external connection terminal **120** may be formed of a solder ball. In another example, the external connection terminal **120** may have a structure including a pillar and a solder. The semiconductor package **1000**A may be mounted on an external substrate, e.g., an interposer or a package substrate, through the external connection terminal **120**.

(24) The through-silicon via (TSV) **130** may penetrate through the substrate **101** in a vertical direction (a Z-axis direction), and may provide an electrical path for connecting the upper pad **105** and the lower pads **104** to each other. The TSV **130** may include a conductive plug and a barrier layer surrounding the conductive plug. The conductive plug may include a metal material, e.g., tungsten (W), titanium (Ti), aluminum (Al), or copper (Cu). The conductive plug may be formed by, e.g., a plating process, a physical vapor deposition (PVD) process, or a chemical vapor deposition (CVD) process. The barrier layer may include an insulating barrier layer or a conductive barrier layer. The insulating barrier layer may be formed of, e.g., oxide, nitride carbide, a polymer, or combinations thereof. The conductive barrier layer may be disposed between the insulating

barrier layer and the conductive plug. For example, the conductive barrier layer may include a metal compound, e.g., tungsten nitride (WN), titanium nitride (TiN), or tantalum nitride (TaN). The barrier layer may be formed in, e.g., a PVD process or a CVD process.
(25) The semiconductor chip **200** may be stacked on the base chip **100**, and may include a substrate **201**, a device layer **210**, and a bump **220**. In the drawing, only one semiconductor chip **200** is illustrated, but the number of semiconductor chips is not limited thereto. For example, two or more semiconductor chips may be stacked on the base chip **100**. The substrate **201** may have characteristics similar to those described for the substrate **101** of the base chip **100**.
(26) The device layer **210** may include a plurality of memory devices. For example, the device layer **210** may include volatile memory devices, e.g., DRAM and SRAM, or nonvolatile memory devices, e.g., PRAM, MRAM, FeRAM, and RRAM. For example, in the semiconductor package **1000**A, the semiconductor chip **200** may include DRAM devices in the device layer **210**. Accordingly, the semiconductor package **1000**A may be used for a high bandwidth memory (HBM) product, an electro data processing (EDP) product, or the like.
(27) The device layer **210** may include a multilayer interconnection layer therebelow. The multilayer interconnection layer may have characteristics similar to those described for the multilayer interconnection layer **112** of the device layer **110** in the base chip **100**. Therefore, devices of the device layer **210** may be electrically connected to the bump **220** through the multilayer interconnection layer. In an example, the base chip **100** may include a plurality of logic devices or memory devices in the device layer **110**, and may be referred to as a buffer chip, a control chip, or the like, according to a function thereof, whereas the semiconductor chip **200** may include a plurality of memory devices in the device layer **210**, and may be referred to as a core chip.
(28) The bump **220** may be disposed on a connection pad **204** on a lower surface of the device layer **210**, and may be connected to devices of the device layer **210** through an interconnection of the multilayer interconnection layer. The bump **220** may electrically connect the semiconductor chip **200** and the base chip **100** to each other. For example, the bump **220** may include a solder or both a pillar and a solder. The pillar may have a cylindrical column shape, or a polygonal column shape, e.g., a square column shape or an octagonal column shape, and may include, e.g., nickel (Ni), copper (Cu), palladium (Pd), platinum (Pt), gold (Au), or combinations thereof. A solder may have a spherical or ball shape, and may include, e.g., tin (Sn), indium (In), bismuth (Bi), antimony (Sb), copper (Cu), silver (Ag), zinc (Zn), lead (Pb), or alloys thereof. The alloys may include, e.g., Sn—Pb, Sn—Ag, Sn—Au, Sn—Cu, Sn—Bi, Sn—Zn, Sn—Ag—Cu, Sn—Ag—Bi, Sn—Ag—Zn, Sn—Cu—Bi, Sn—Cu—Zn, Sn—Bi—Zn, and the like.
(29) The adhesive film **300** may be disposed between the base chip **100** and the semiconductor chip **200** to surround a side surface of the bump **220**, and may fix the semiconductor chip **200** to the base chip **100**. For example, as illustrated in FIG. **1**A, the adhesive film **300** may be disposed between a top surface of the base chip **100** and a bottom surface of the semiconductor chip **200** to completely fill an empty spaced therebetween (e.g., while completely surrounding a perimeter of each of the bumps **220**). As illustrated in FIGS. **1**A to **1**C, the adhesive film **300** may protrude outwardly from, e.g., beyond, a side surface **200**S of the semiconductor chip **200** and may cover at least a portion of the side surface **200**S of the semiconductor chip **200**, e.g., the adhesive film **300** may protrude outwardly beyond the side surface **200**S of the semiconductor chip **200** around an entire perimeter of the semiconductor chip **200** (FIG. **1**C).
(30) The adhesive film **300** may be a non-conductive film (NCF), and may include, e.g., any type of polymer film which may perform a thermocompression bonding process. The thermocompression bonding process may allow at least a portion of the adhesive film **300** to form an extension region flowing outwardly of the semiconductor chip **200** and protruding farther than the side surface **200**S of the semiconductor chip **200**. In the present disclosure, the flow region or the expansion region of the adhesive film **300** may be limited using a dam structure **500** to be

described later. Accordingly, the adhesive film **300** may protrude farther than the side surface **200**S of the semiconductor chip **200** to be in, e.g., direct, contact with internal surfaces **500**IS of the dam structures **500**. For example, the adhesive film **300** may not be in contact with the upper surfaces **500**US of the dam structures **500**, but example embodiments are not limited thereto, e.g., the adhesive film **300** may be in contact with upper surfaces **500**US of at least some of the dam structures **500**. It is noted that the upper surfaces **500**US of the dam structures **500** refer to surfaces in the X-Y plane that face away from the base chip **100**.

(31) The encapsulant **400** may be disposed on the base chip **100**, and may encapsulate at least a portion of the base chip **100**, the semiconductor chip **200**, the adhesive film **300**, and the dam structure **500**. For example, as illustrated in FIG. **1**A, the encapsulant **400** may have a predetermined thickness and may cover an upper surface of the semiconductor chip **200**. In another example, as illustrated in FIG. **4**A, the encapsulant **400** may not cover the upper surface of the semiconductor chip **200**, so the upper surface of the semiconductor chip **200** may be exposed from the encapsulant **400**. The encapsulant **400** may include, e.g., an epoxy mold compound (EMC).

(32) The dam structure **500** may be disposed on the base chip **100**, and may include a seed layer **501** disposed on an upper surface **100**US of the base chip **100** and a metal layer **502** disposed on the seed layer **501**. For example, the seed layer **501** may have a single-layer structure or a multilayer structure including a metallic material or a material exhibiting metallic characteristics, e.g., a material including at least one of copper (Cu), aluminum (Al), nickel (Ni), silver (Ag), gold (Au), platinum (Pt), tin (Sn), lead (Pb), titanium (Ti), chromium (Cr), palladium (Pd), indium (In), zinc (Zn), carbon (C), or an alloy including at least one metal or two or more metals thereof. The metal layer **502** may be a plating layer formed by an electrolytic plating process using the seed layer **501**. For example, Referring to FIG. **1**B, the dam structures **500** may include the inner surfaces **500**IS facing the semiconductor chip **200**, outer surfaces **500**OS opposing the inner surfaces **500**IS, and the upper surfaces **500**US connecting the inner surfaces **500**IS and the outer surfaces **500**OS to each other. The dam structures **500** may also have lateral surfaces perpendicular to the upper surfaces **500**US and the inner surfaces **500**IS, e.g., the lateral surfaces may face corner portions **100**CR of the base chip **100** (FIG. **1**C). At least a portion of the inner surfaces **500**IS may be in contact with the adhesive film **300**, and at least a portion of the upper surfaces **500**US and the outer surfaces **500**OS, e.g., and the lateral surfaces, may be in contact with the encapsulant **400**.

(33) The dam structure **500** may be formed to have a predetermined height to avoid contact with a bonding head (see ‘20’ of FIG. **7**E) in a thermal compression process of the semiconductor chip **200** and to effectively limit the extension region of the adhesive film **300**. As an example, the upper surfaces **500**US of the dam structures **500** may be disposed on a level lower than a level of the upper surface **200**US of the semiconductor chip **200**, e.g., relative to the base chip **100**. For example, the dam structures **500** may have a height **500**H lower than a height from the upper surface **100**US of the base chip **100** to the upper surface **200**US of the semiconductor chip **200**. The height **500**H of the dam structures **500** may be less than about 100% of the height from the upper surface **100**US of the base chip **100** to the upper surface **200**US of the semiconductor chip **200**. For example, the height **500**H of the dam structures **500** may be about 90% or less to about 50% or more, about 90% or less to about 60% or more, or about 90% or less to about 70% or more of the height from the upper surface **100**US of the base chip **100** to the upper surface **200**US of the semiconductor chip **200**. When the height **500**H of the dam structures **500** is less than about 50% of the height from the upper surface **100**US of the base chip **100** to the upper surface **200**US of the semiconductor chip **200**, the adhesive film **300** may overflow. When the height **500**H of the dam structures **500** is greater than about 90% of the height from the upper surface **100**US of the base chip **100** to the upper surface **200**US of the semiconductor chip **200**, the bonding head (see ‘20’ of FIG. **7**E) may be brought into contact with the dam structures **500** during the thermocompression bonding process, and thus, sufficient pressure may not be applied to the semiconductor chip **200**.

(34) The dam structures **500** may be formed to surround the side surface **200**S of the semiconductor

chip **200**. For example, as illustrated in FIG. **1**C, the dam structures **500** may include dam structures **500-1**, **500-2**, **500-3**, and **500-4**, respectively corresponding to side surfaces **200**S**1**, **200**S**2**, **200**S**3**, and **200**S**4** of the semiconductor chip **200**, on a plane (an X-Y plane). The dam structures **500** may have a shape extending continuously (see FIG. **1**C) or discontinuously (to be described with reference to FIG. **3**) between the side surfaces **200**S of the semiconductor chip **200** and the side surfaces **100**S of the base chip **100**, e.g., as viewed in a plan view. The dam structures **500** may be spaced apart from each other in the corner portion **100**CR of the base chip **100** to secure fluidity of the encapsulant **400**, and a space between the dam structures **500** spaced apart from each other may be filled with the encapsulant **400**. As an example, the dam structures **500** may extend to have a length of about 90% or less of lengths of the side surfaces **100**S of the corresponding base chip **100**, respectively. For example, the first dam structure **500-1** may extend to have a first length **500**SL**1** of about 90% or less of the length **100**SL**1** of the first side surface **100**S**1** of the corresponding base chip **100**. In addition, the second dam structure **500-2** may extend to have a second length **500**SL**2** of about 90% or less of the length **100**SL**2** of the second side surface **100**S**2** of the corresponding base chip **100**. When the length of the dam structures **500** exceeds about 90% of the length of the side surfaces **100**S of the base chip **100**, the fluidity of the encapsulant **400** may be reduced, so that spaces between the dam structures **5000** and the semiconductor chip **200** may not be filled with the encapsulant **400**. For example, in FIG. **1**C, the adhesive film **300** is illustrated as being in continuous contact with inner surfaces of the dam structures **500** between the side surface **200**S of the semiconductor chip **200** and the dam structures **500**. In another example, contact surfaces between the adhesive film **300** and the dam structures **500** may be discontinuously formed.
(35) The dam structures **500** may be spaced apart from the side surfaces **200**S of the semiconductor chip **200** by a first distance d1 to secure an extension region or a flow region of the adhesive film **300**. The first distance d1 may be selected in consideration of temperature and pressure conditions of the thermocompression bonding process, fluidity of the adhesive film **300**, and the like. The dam structures **500** may have a predetermined separation distance from the side surfaces **100**S of the base chip **100** to be prevented from overlapping a scribe lane (see 'SL' of FIG. **7**I) of the base chip **100**. For example, as illustrated in FIG. **1**C, the dam structures **500-1**, **500-2**, **500-3**, and **500-4** may be spaced part from the side surfaces **100**S**1**, **100**S**2**, **100**S**3**, and **100**S**4** of the corresponding base chip **100** by a second distance d2, respectively. The second distance d2 may be about 5 μm or more, or about 10 μm or more. When the second distance d2 is less than about 5 μm, the dam structures **500** are exposed outwardly of the encapsulant **400** in a cutting process of the base chip **100**, resulting in poor exterior and deteriorated reliability. For example, the second distance d2 may be adjusted in inverse proportion to the first distance d1, but may be secured as the above-described separation distance from the side surfaces **100**S of the base chip **100** to prevent exposure of the dam structure **500**.
(36) FIG. **2** is a partially enlarged view illustrating a modified example of the semiconductor package of FIG. **1**A. FIG. **2** illustrates a region corresponding to FIG. **1**B of a semiconductor package **1000**Aa according to the modified example.
(37) Referring to FIG. **2**, in the modified example, the upper surfaces **500**US of the dam structures **500** may be, e.g., at least partially, covered with the adhesive film **300**. For example, the dam structures **500** may have the inner surfaces **500**IS facing the semiconductor chip **200**, the outer surfaces **500**OS opposing the inner surfaces **200**IS, and the upper surfaces **500**US connecting the inner surfaces **500**IS and the outer surfaces **500**OS to each other. At least some of the inner surfaces **500**IS and the upper surfaces **500**US may be in contact with the adhesive film **300**, and some of the remaining upper surfaces **500**US and the outer surfaces **500**OS may be in contact with the encapsulant **400**. Even in this case, the contact surfaces between the upper surfaces **500**US of the dam structures **500** and the adhesive film **300** may not be formed to have a constant width or may not be formed to be constant in a length direction of the dam structures **500**. For example,

according to example embodiments, the adhesive film **300** may have the shape illustrated in FIG. **1**B in some regions and a shape illustrated in FIG. **2** in other regions.
(38) FIG. **3** is a plan view illustrating a modified example of the semiconductor package of FIG. **1**A. FIG. **3** illustrates a region corresponding to FIG. **1**C of a semiconductor package **1000**Ab according to the modified example.
(39) Referring to FIG. **3**, in the modified example, the dam structures **500** may each include a plurality of dam units **500**U arranged in parallel to side surfaces **200**S of a semiconductor chip **200**. In FIGS. **1**C and **3**, the adhesive film **300** is illustrated as being in continuous contact with the inner surface of the dam structures **500** between the side surfaces **200**S of the semiconductor chip **200** and the dam structures **500**. According to example embodiments, contact surfaces between the adhesive film **300** and the dam structures **500** may be discontinuously formed.
(40) In another example, unlike that illustrated in FIGS. **1**C and **3**, the adhesive film **300** may be brought into contact with the dam structures **500** in an extension region having a maximum width in a direction perpendicular to the side surfaces **200**S of the semiconductor chip **200**, and may be spaced apart from the dam structures **500** in the extension region having a width smaller than or equal to the maximum width. For example, the extension region of the adhesive film **300** may not have a uniform width. According to the present embodiment, since the dam structures **500** include a plurality of dam units **500**U spaced apart from each other, a non-contact region between the adhesive film **300** and the side surfaces **200**S of the semiconductor chip **200** may be filled with the encapsulant **400**. In addition, a flow region of the adhesive film **300** between the plurality of dam units **500**U may be additionally secured to prevent overflow. A distance d3 between, e.g., adjacent ones of, the plurality of dam units **500**U may be within a range of about 5 μm or less, e.g., about 5 μm to about 1 μm, about 4 μm to about 1 μm, and about 3 μm to about 1 μm. When the distance d3 between the plurality of dam units **500**U exceeds about 5 μm, the adhesive film **300** may pass between adjacent ones of the plurality of dam units **500**U to protrude farther outwardly than the dam structures **500**. When the distance d3 between the plurality of dam units **500**U is less than about 1 μm, filling properties of the encapsulant **400** may be deteriorated.
(41) FIG. **4**A is a cross-sectional view of a semiconductor package according to an example embodiment, and FIG. **4**B is a partially enlarged view of region “B” of FIG. **4**A.
(42) Referring to FIGS. **4**A and **4**B, a semiconductor package **1000**B according to an example embodiment may have the same or similar features to those of the semiconductor package **1000**A illustrated in FIG. **1**A, except that a plurality of semiconductor chips **200**A, **200**B**1**, **200**B**2**, and **200**C may be stacked on the base chip **100** in a first direction (the Z-axis direction). Therefore, descriptions overlapping with those described with reference to FIGS. **1**A and **1**B will be omitted.
(43) For example, as illustrated in FIGS. **4**A-**4**B, the semiconductor package **1000**B according to the present embodiment may further include a first semiconductor chip **200**A mounted on the base chip **100**, second semiconductor chips **200**B**1**, **200**B**2**, and **200**C sequentially stacked on the first semiconductor chip **200**A, and second adhesive films **300**B**1**, **300**B**2**, and **300**C, respectively disposed below the second semiconductor chips **200**B**1**, **200**B**2**, and **200**C. The first to second semiconductor chips **200**A, **200**B**1**, **200**B**2**, and **200**C stacked on the base chip **100** may include a through-silicon via (TSV) **230**, and may be electrically connected to each other by the TSV **230**. In an example, the number of semiconductor chips **200** stacked on the base chip **100** may be two, three, or five or more. Among the second semiconductor chips **200**B**1**, **200**B**2** and **200**C, the uppermost second semiconductor chip **200**C does not include a TSV, and an upper surface thereof may be exposed from the encapsulant **400**. According to example embodiments, an upper surface of the uppermost second semiconductor chip **200**C may be covered with the encapsulant **400**.
(44) According to the present embodiment, damage to the base chip **100** may be prevented in a process of cutting extension regions of the second adhesive films **300**B**1**, **300**B**2**, and **300**C using the dam structures **500**. As a result, reliability of the semiconductor package **1000**B may be improved. For example, the second adhesive films **300**B**1**, **300**B**2**, and **300**C disposed between the

first semiconductor chip **200**A and the lowermost second semiconductor chip **200**B**1** and between the second semiconductor chips **200**B**1**, **200**B**2**, and **200**C may be formed such that the side surfaces **300**BS**1**, **300**BS**2**, and **300**CS thereof are disposed between the inner surface **500**IS and the outer surface **500**OS of the dam structures **500**, e.g., along the X-axis direction (FIG. **4**A) or on the upper surface **500**US. In addition, as described with reference to FIG. **1**A, or the like, the dam structures **500** may be disposed on the base chip **100** to surround the periphery of the first semiconductor chip **200**A, and the expansion region or a flow region of the first adhesive film **300**A disposed between the base chip **100** and the first semiconductor chip **200**A may be limited, so that poor exterior and a decrease in reliability caused by the first adhesive film **300**A may be prevented. Accordingly, at least a portion of the first adhesive film **300**A may fill a space between the first semiconductor chip **200**A and the dam structures **500** and may be in contact with the inner surface **500**IS of the dam structures **500**, and at least a portion of the adhesive films **300**B**1**, **300**B**2**, and **300**C may overlap the dam structures **500** in the first direction (the Z-axis direction). In addition, at least a portion of the second adhesive films **300**B**1**, **300**B**2**, and **300**C may be in contact with the upper surface **500**US of the dam structures **500**. For example, the lowermost second adhesive film **300**B**1** may be in, e.g., direct, contact with the upper surfaces **500**US of the dam structures **500**. In addition, the second adhesive films **300**B**1**, **300**B**2**, and **300**C may be formed to have substantially the same width, e.g., along the X-axis direction, by a cutting process, and side surfaces **300**BS**1**, **300**BS**2**, and **300**CS thereof may be substantially coplanar with each other. For example, a maximum width Wa of the first adhesive film **300**A in a second direction (the X-axis direction) may be less than or equal to a maximum width Wb of the second adhesive films **300**B**1**, **300**B**2**, and **300**C in the second direction (the X-axis direction).

(45) FIG. **5**A is a plan view of a semiconductor package according to an example embodiment, and FIG. **5**B is a cross-sectional view taken along line I-I′ of FIG. **5**A.

(46) Referring to FIGS. **5**A and **5**B, a semiconductor package **10000**A according to an example embodiment may include a package substrate **600**, an interposer substrate **700**, and at least one chip structure **1000**. The semiconductor package **10000**A may further include a logic chip or a processor chip **800** disposed to be adjacent to the chip structure **1000** on the interposer substrate **700**.

(47) The package substrate **600** may include a lower pad **612** disposed on a lower surface of a body, an upper pad **611** disposed on an upper surface of the body, and a redistribution circuit **613** electrically connecting the lower pad **612** and the upper pad **611** to each other. The package substrate **600** may be a support substrate on which the interposer substrate **700**, the processor chip **800**, and the chip structure **1000** are mounted, and may be a substrate for a semiconductor package including a printed circuit board (PCB), a ceramic substrate, a glass substrate, a tape interconnection board, or the like. The body of the package substrate **600** may include different materials depending on the type of substrates. For example, when the package substrate **600** is a PCB, it may have a form in which an interconnection layer is additionally laminated on one side or both sides of a body copper clad laminate or a copper clad laminate. A solder resist layer may be formed on each of the lower and upper surfaces of the package substrate **600**. The lower and upper pads **612** and **611** and the redistribution circuit **613** may form an electrical path connecting the lower surface and the upper surface of the package substrate **600** to each other.

(48) For example, the lower and upper pads **612** and **611** and the redistribution circuit **513** may be formed of a metallic material or a material exhibiting metallic characteristics, e.g., a material including at least one of copper (Cu), aluminum (Al), nickel (Ni), silver (Ag), gold (Au), platinum (Pt), tin (Sn), lead (Pb), titanium (Ti), chromium (Cr), palladium (Pd), indium (In), zinc (Zn), carbon (C), or an alloy including at least one metal or two or more metals thereof. The redistribution circuit **613** may include multiple redistribution layers and vias connecting the multiple redistribution layers to each other. An external connection terminal **620**, connected to the lower pad **512**, may be disposed on the lower surface of the package substrate **600**. The external connection terminal **620** may include, e.g., tin (Sn), indium (In), bismuth (Bi), antimony (Sb),

copper (Cu), silver (Ag), zinc (Zn), lead (Pb), and/or alloys thereof.

(49) The interposer substrate **700** may include a substrate **701**, a lower passivation layer **703**, a lower pads **705**, an interconnection layer **710**, bumps **720**, and a through-electrodes **730**. The chip structure **1000** and the processor chip **800** may be stacked on the package substrate **600** via the interposer substrate **700**. The interposer substrate **700** may electrically connect the chip structure **1000** and the processor chip **800** to each other.

(50) The substrate **701** may be formed of, e.g., silicon, an organic material, plastic, or a glass substrate. When the substrate **701** is a silicon substrate, the interposer substrate **700** may be referred to as a silicon interposer. In addition, when the substrate **701** is an organic substrate, the interposer substrate **700** may be referred to as a panel interposer.

(51) The lower passivation layer **703** may be disposed on a lower surface of the substrate **701**, and the lower pads **705** may be on the lower passivation layer **703**. The lower pads may be connected to the through-electrodes **730**. The chip structure **1000** and the processor chip **800** may be electrically connected to the package substrate **600** through the bumps **720** disposed on the lower pads **705**.

(52) The interconnection layer **710** may be disposed on an upper surface of the substrate **701**, and may include an interlayer insulating layer **711** and a single-layer or multilayer interconnection structure **712**. When the interconnection layer **710** has a multilayer interconnection structure, interconnections of different layers may be connected to each other through vertical contacts.

(53) The through-electrode **730** may extend from the upper surface to the lower surface of the substrate **701** to penetrate through the substrate **701**. In addition, the through-electrode **730** may extend inwardly of the interconnection layer **710** to be electrically connected to interconnections of the interconnection layer **710**. When the substrate **701** is silicon, the through electrode **730** may be referred to as a TSV. Other structures and materials of the through-electrode **730** are the same as those described for the semiconductor package **1000**A of FIG. **1**A. According to example embodiments, the interposer substrate **700** may include only an interconnection layer therein, or may not include a through-electrode.

(54) The interposer substrate **700** may be used to convert or transmit an input electrical signal between the package substrate **600** and the chip structure **1000** or the processor chip **800**. Accordingly, the interposer substrate **700** may not include devices, e.g., active devices or passive devices. According to example embodiments, the interconnection layer **710** may also be disposed below the through-electrode **730**. For example, a positional relationship between the interconnection layer **710** and the through electrode **730** may be relative.

(55) The bumps **720** may be disposed on a lower surface of the interposer substrate **700** and may be electrically connected to an interconnection of the interconnection layer **710**. The interposer substrate **700** may be stacked on the package substrate **600** through the bumps **720**. The bumps **720** may be connected to the lower pads **705** through the interconnections of the interconnection layer **710** and the through-electrodes **730**. In an example, some pads **705** used for power or ground, among the lower pads **705**, may be integrated and connected to the bumps **720**, so that the number of the lower pads **705** may be greater than the number of the bumps **720**.

(56) The logic chip or the processor chip **800** may include, e.g., a central processor (CPU), a graphics processor (GPU), a field programmable gate array (FPGA), a digital signal processor (DSP), a cryptographic processor, a microprocessor, a microcontroller, an analog-to-digital converter, an application-specific IC (ASIC), or the like. According to the types of devices included in the logic chip **800**, the semiconductor package **10000** may be classified into a server-oriented semiconductor package or a mobile-oriented semiconductor package.

(57) The chip structure **1000** may have characteristics that are the same as or similar to those of the semiconductor packages **1000**A, **1000**Aa, **1000**Ab, and **1000**B illustrated in FIG. **1**A to **4**B. For example, the chip structure **1000** may include the dam structures **500** disposed on the base chip **100**, and the adhesive films **300**A, **300**B**1**, **300**B**2**, and **300**C may be formed to have specific shapes based on the dam structures **500**.

(58) The semiconductor package **10000**A according to an example embodiment may further include an inner encapsulant covering a side surface and an upper surface of the chip structure **1000** and the processor chip **800** on the interposer substrate **700**. The semiconductor package **10000**A may further include an external encapsulant covering the interposer substrate **700** and the internal encapsulant on the package substrate **600**. According to example embodiments, the external encapsulant and the internal encapsulant may be formed together and may not be distinguished from each other. According to example embodiments, the internal encapsulant may cover only the upper surface of the processor chip **800** and may not cover the upper surface of the chip structure **1000**.
(59) FIG. **6** is a cross-sectional view of a semiconductor package according to an example embodiment.
(60) Referring to FIG. **6**, a semiconductor package **10000**B according to an example embodiment may be understood to have features that are the same as or similar to those of the semiconductor packages illustrated in FIGS. **1**A to **4**B, except that a logic chip (i.e., a logic chip **800**) and the plurality of semiconductor chips **200**A, **200**B**1**, **200**B**2**, and **200**C are sequentially stacked on the base chip **100**. Therefore, descriptions overlapping with those described with reference to FIGS. **1**A to **4**B will be omitted.
(61) The plurality of semiconductor chips **200**A, **200**B**1**, **200**B**2**, and **200**C may include a memory chip, may be stacked on the logic chip **800** in a vertical direction (the Z-axis direction), and may be electrically connected to the package substrate **600** through a through-electrode **830** in the logic chip **800**. According to example embodiments, the plurality of semiconductor chips **200**A, **200**B**1**, **200**B**2**, and **200**C may be disposed side by side in a horizontal direction (the X-axis and Y-axis directions) on an upper surface of the logic chip **800**.
(62) The logic chip **800** may include elements similar to the base chip **100** illustrated in FIG. **1**A, or the like. For example, the logic chip **800** may include a substrate **801**, a device layer **810**, an external connection terminal **820**, the through-electrode **830**, and the like. As described with reference to FIGS. **5**A and **5**B, the logic chip **800** may include a processor chip, e.g., a CPU or a GPU.
(63) FIGS. **7**A to **7**I are cross-sectional views of stages in a method of manufacturing a semiconductor package according to an example embodiment.
(64) Referring to FIG. **7**A, a preliminary seed layer **501***p* may be formed on the base chip **100**. The base chip **100** may be a semiconductor wafer **100**W including a plurality of base chips **100** divided by scribe lanes SL.
(65) In detail, the semiconductor wafer **100**W may be disposed on a carrier **10**. The carrier **10** may include a support substrate **11** and an adhesive material layer **12**. The semiconductor wafer **100**W may be attached to the carrier **10** such that a lower surface of the base chip **100**, on which the external connection terminal **120** is disposed, is directed toward the adhesive material layer **12**. The external connection terminal **120** may be covered with the adhesive material layer **12**, and a lower surface of the semiconductor wafer **100**W may be in contact with an upper surface of the adhesive material layer **12**.
(66) The preliminary seed layer **501***p* may be conformally formed along the upper surface **100**US of the base chip **100** using a deposition process, e.g., a CVD or PVD process. The preliminary seed layer **501***p* may include, e.g., titanium (Ti), copper (Cu), or the like, and may have etching selectivity with respect to the upper pad **105**.
(67) Referring to FIG. **7**B, a photosensitive material layer PR may be formed on the preliminary seed layer **501***p*. The photosensitive material layer PR may be a positive or negative type photoresist. The photosensitive material layer PR may be formed to have a thickness sufficient to secure heights of the dam structures **500** described with reference to FIGS. **1**A to **1**C. The photosensitive material layer PR may be formed by coating a photosensitive ink on the semiconductor wafer **100**W.

(68) Referring to FIG. **7**C, the photosensitive material layer PR may be patterned such that trenches **502***p* are formed to expose the preliminary seed layer **501***p*. The photosensitive material layer PR may be patterned using a photolithography process. The photolithography process may include a series of processes, e.g., an exposure process using a photomask, a developing process, a cleaning process, and the like. The trenches **502***p* may be formed to surround the upper pads **105** of the semiconductor wafer **100**W. For example, referring to FIG. **7**C, a single trench **502***p* may surround all the upper pads **105** of one base chip **100**, as viewed in a top view (e.g., two trenches **502***p* in a cross-sectional view correspond to a ring-shaped structure). Also, the trenches **502***p* may be formed to be spaced apart from an outermost upper pad **105** by a predetermined distance.
(69) Referring to FIG. **7**D, the dam structures **500** including the seed layer **501** and the metal layer **502** may be formed. The metal layer **502** may be formed by performing an electroplating process using the preliminary seed layer **501***p* of FIG. **7**C, and may be formed of a metal, e.g., copper (Cu) or a copper (Cu) alloy. The seed layer **501** may be formed by forming the metal layer **502**, removing the photosensitive material layer PR of FIG. **7**C, and etching the preliminary seed layer **501***p* of FIG. **7**C exposed from the metal layer **502**.
(70) Referring to FIG. **7**E, the first semiconductor chips **200**A may be mounted on the base chip **100** by performing a thermocompression bonding process. The first adhesive film **300**A may be disposed below the first semiconductor chip **200**A. The first semiconductor chip **200**A may be vacuum-adsorbed to a bonding head **20** to be picked and placed on the semiconductor wafer **100**W. The dam structures **500** may have a predetermined separation distance d4 from a lower surface of the bonding head **20**. If the dam structures **500** were to be in contact with the lower surface of the bonding head **20**, sufficient pressure could not have been transferred in the thermocompression bonding process of the first semiconductor chip **200**A, and adhesive force of the first semiconductor chip **200**A would have been reduced.
(71) Referring to FIG. **7**F, the plurality of second semiconductor chips **200**B**1**, **200**B**2**, and **200**C may be sequentially stacked on the first semiconductor chip **200**A by repeatedly performing the thermocompression bonding process. Second adhesive films **300**B**1**, **300**B**2**, and **300**C may be disposed below the plurality of second semiconductor chips **200**B**1**, **200**B**2**, and **200**C, respectively. The second adhesive films **300**B**1**, **300**B**2**, and **300**C may have an extension region EX protruding farther outwardly than the dam structures **500** by a thermocompression bonding process.
(72) Referring to FIGS. **7**F and **7**G, the extension region EX of the second adhesive films **300**B**1**, **300**B**2**, and **300**C of FIG. **7**F may be cut, so that side surfaces **300**BS**1** and **300**BS**2** of the second adhesive films **300**B**1**, **300**B**2**, and **300**C may be formed to be disposed on upper surfaces of the dam structures **500**.
(73) That is, since the extension region EX may cause poor exterior and a decrease in reliability of the semiconductor package, the extension region EX of the second adhesive films **300**B**1**, **300**B**2**, and **300**C may be removed, according to example embodiments, without damage to the base chip **100** using the dam structures **500**. Due to the cutting process, the side surfaces **300**BS**1**, **300**BS**2**, **300**C of the second adhesive films **300**B**1**, **300**B**2**, **300**C may be substantially coplanar, and the second adhesive films **300**B**1**, **300**B**2**, and **300**C may not protrude further outwardly than the dam structures **500**.
(74) Referring to FIG. **7**H, the encapsulant **400** may be formed on the semiconductor wafer **100**W, and an upper surface of the encapsulant **400** may be planarized using a polishing apparatus **30**. In an example embodiment, the encapsulant **400** may be formed to cover the side surfaces **300**BS**1**, **300**BS**2**, and **300**C of the second adhesive films **300**B**1**, **300**B**2**, and **300**C and side surfaces and at least a portion of upper surfaces of the dam structures **500**. Due to the planarization process, an upper surface of the encapsulant **400** may be disposed to be substantially coplanar with an upper surface of the uppermost second semiconductor chip **200**C. The planarization process may be, e.g., a chemical mechanical polishing (CMP) process.
(75) Referring to FIG. **7**I, the encapsulant **400** and the semiconductor wafer **100**W may be cut

along scribe lanes SL to divide a plurality of semiconductor packages **1000′**. The plurality of semiconductor packages **1000′** may have features that are the same as or similar to those of the semiconductor packages described with reference to FIGS. **1**A to **4**B. The semiconductor package **1000′**, completed through the above-described manufacturing process, may have reliability improved by controlling the expansion region of the first and second adhesive films **300**A, **300**B**1**, **300**B**2**, and **300**C using the dam structures **500**.
(76) By way of summation and review, example embodiments provide a semiconductor package having improved reliability. That is, according to example embodiments, a semiconductor package having improved reliability may be provided by introducing a dam structure controlling an extension region of an adhesive film.
(77) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor package, comprising: a base chip; a first semiconductor chip on the base chip; at least one second semiconductor chip on the first semiconductor chip in a first direction perpendicular to an upper surface of the base chip, each of the first semiconductor chip and the at least one second semiconductor chip including a through-silicon via (TSV), and the first semiconductor chip and the at least one second semiconductor chip being electrically connected to each other via the TSV; dam structures on the base chip, the dam structures surrounding a periphery of the first semiconductor chip; a first adhesive film between the base chip and the first semiconductor chip, the first adhesive film including a first portion filling a space between the first semiconductor chip and the base chip and a second portion disposed in a space between a side of the first semiconductor chip and each of the dam structures; a second adhesive film between the first semiconductor chip and the at least one second semiconductor chip, at least a portion of the second adhesive film overlapping an upper surface of each of the dam structures in the first direction; and an encapsulant encapsulating at least a portion of each of the dam structures, the first semiconductor chip, and the at least one second semiconductor chip.
2. The semiconductor package as claimed in claim 1, wherein a maximum width of the first adhesive film in a second direction, perpendicular to the first direction, is less than or equal to a maximum width of the second adhesive film in the second direction.
3. The semiconductor package as claimed in claim 1, wherein: the dam structures have inner surfaces facing the first semiconductor chip, outer surfaces opposing the inner surfaces, and upper surfaces connecting the inner surfaces and the outer surfaces to each other; and at least a portion of the upper surfaces and the outer surfaces of the dam structures are in contact with the encapsulant.
4. The semiconductor package as claimed in claim 3, wherein the second portion of the first adhesive film filling the space between the first semiconductor chip and the dam structures is in contact with the inner surfaces of the dam structures.
5. The semiconductor package as claimed in claim 3, wherein side surfaces of the second adhesive film are on the upper surfaces of the dam structures.
6. The semiconductor package as claimed in claim 5, wherein: the second adhesive film is provided in a plurality of second adhesive films, the at least one second semiconductor chip includes a

plurality of second semiconductor chips stacked on top of each other along the first direction, the plurality of second adhesive films being positioned between adjacent ones of the plurality of second semiconductor chips, respectively, and side surfaces of the plurality of second adhesive films are coplanar with each other.
7. The semiconductor package as claimed in claim 1, wherein the upper surface of each of the dam structures is positioned between an upper surface of the first semiconductor chip and a lower surface of the first semiconductor chip, and wherein the upper surface of each of the dam structures is lower than the upper surface of the first semiconductor chip.
8. The semiconductor package as claimed in claim 1, wherein the dam structures have a height of about 90% or less and about 60% or more of a height from the upper surface of the base chip to an upper surface of the first semiconductor chip.
9. The semiconductor package as claimed in claim 1, wherein the dam structures extend along corresponding side surfaces of the first semiconductor chip, the dam structures being respectively spaced apart from the corresponding side surfaces of the first semiconductor chip.
10. The semiconductor package as claimed in claim 9, wherein the dam structures have a predetermined separation distance from side surfaces of the base chip.
11. The semiconductor package as claimed in claim 10, wherein the predetermined separation distance is about 10 micrometers or more.
12. The semiconductor package as claimed in claim 1, wherein, in a plan view, the dam structures respectively correspond to side surfaces of the first semiconductor chip and have a shape extending continuously or discontinuously between the side surfaces of the first semiconductor chip and side surfaces of the base chip.
13. The semiconductor package as claimed in claim 12, wherein the dam structures have a length of about 90% or less of a length of the side surfaces of the base chip.
14. The semiconductor package as claimed in claim 12, wherein the dam structures are spaced apart from each other in a corner portion of the base chip, a space between the dam structures being filled with the encapsulant.
15. The semiconductor package as claimed in claim 1, wherein the dam structures include a seed layer on the upper surface of the base chip, and a metal layer on the seed layer.
16. A semiconductor package, comprising: a base chip; a semiconductor chip on the base chip; dam structures on the base chip and surrounding a side surface of the semiconductor chip, an upper surface of each of the dam structures being at a level lower than a level of an upper surface of the semiconductor chip; and an adhesive film having a first portion between the base chip and the semiconductor chip and a second portion protruding beyond the side surface of the semiconductor chip to be in contact with an inner surface of the dam structures, wherein an upper surface of the second portion contacts the side surface of the semiconductor chip, wherein the second portion of the adhesive film is disposed in a space between a side of first semiconductor chip and each of the dam structures, and wherein the upper surface of each of the dam structures is positioned between the upper surface of the semiconductor chip and a lower surface of the semiconductor chip.
17. The semiconductor package as claimed in claim 16, wherein the adhesive film is not in contact with the upper surface of the dam structures.
18. A semiconductor package, comprising: a base chip; a first semiconductor chip and second semiconductor chips sequentially stacked on the base chip; dam structures on the base chip, the dam structures respectively corresponding to side surfaces of the first semiconductor chip; a first adhesive film including a first portion disposed between the base chip and the first semiconductor chip and a second portion disposed in a space between a side of the first semiconductor chip and each of the dam structures; and second adhesive films between the first semiconductor chip and a lowermost second semiconductor chip of the second semiconductor chips, and between adjacent ones of the second semiconductor chips, wherein the first adhesive film or a lowermost second adhesive film of the second adhesive films is in contact with an upper surface of each of the dam

structures, wherein an uppermost second adhesive film of the second adhesive films is disposed between an uppermost second semiconductor chip of the second semiconductor chips and another second semiconductor chip immediately below the uppermost second semiconductor chip, and wherein the uppermost second adhesive film has a portion protruding beyond a side surface of the uppermost second semiconductor chip and overlapping the upper surface of each of the dam structures.

19. The semiconductor package as claimed in claim 18, wherein each of the dam structures includes a plurality of dam units arranged side by side with the side surfaces of the first semiconductor chip.

20. The semiconductor package as claimed in claim 19, wherein a distance between adjacent ones of the plurality of dam units is about 5 micrometers or less.